

GENERAL RISK ASSESSMENT TEMPLATE

Work area / operation	CB11.10.403	Assessor's name	Hannah Murphy, Maryam Qutub, Yara Haddad
Other persons consulted	Katia Stephan	Date of safety assessment	26/08/2026
Subject Coordinator's Name	Gavin Paul	Lab Supervisor's Name	Felipe Ulloa Gutierrez

ACTIVITY - Describe hazardous activities related to the work area or operation.	ASSOCIATED HAZARDS	INHERENT RISK - Harm that could occur from these hazards if controls fail or are not in place.	EXISTING CONTROL MEASURES	PROPOSED CONTROL MEASURES - Proposed action to minimise risk to an acceptable level.	TARGET DATE - To implement proposed controls	RESIDUAL RISK LEVEL (H,M,L)
Electrical cables mishandled	Can be a trip hazard. Damaged or exposed cables. Damaged robot parts.	- Trip, fall or slip suffering an injury. - Electrical shock - Short circuit	- Organised cables that are not in walking path. - Inspecting cables for existing damage before usage. - Handling wires and cables safely.	- Secure free wires route cables away from walking path using cable covers or ties. - Replace any damaged cables before using.	Before commencing robot operation. (07/10/2026)	L
Leaving the robot arm unattended	Robot may move unexpectedly causing collisions with nearby objects or people. Unauthorized usage	- Robot arm collisions could cause injuries - Damage robot parts and surrounding environment.	- Emergency stop button is accessible - Operator should be present when robot is used. - Access to robot workspace is restricted. - Induction before robot usage.	- Stop robot if no able to continue monitoring it. - Ensure no object block the path of the robot. - Ensure people are aware that the robot is being used.	Before and during robot operation. (07/10/2026)	L
Using inappropriate load on the robot	Dropped objects, collisions, heavy payloads may cause unexpected motion.	- Heavy loads could fall from robot - Bodily injuries - Damaged robot, surrounding environment and payload object.	- Follow robot restrictions and specifications given by the manufacturer. - Restrict access to robot area when in use of payload. - Safety inductions before using robot with payload.	- Confirm payload is within the specified limit. - Test using slow movements. - Ensure nothing is blocking the pathway of the robot.	Before and during robot operation. (07/10/2026)	L
Robot arm moves incorrectly	Unplanned motion, collision, pinching, crushing.	- Incorrect robot movement could cause bodily injuries. - Damage to equipment and surrounding environment.	- Emergency stop button is accessible - Use low operation speed during testing. - Run simulations prior to using the robot - Workspace is kept clear.	- Verify code through simulation before testing on the robot - Ensure no additional objects or people are in the path of the robot. - Use emergencies stop button immediately when incorrect moves occur.	Before and during robot operation. (07/10/2026)	L

Incorrect manual handling	Heavy robotic parts. Strain or dropping parts	- Incorrect manual handling could cause bodily harm - Dropping heavy parts could cause bodily injuries. - Damaged robot.	- Use the appropriate lifting technique. - Avoid lifting heavy objects alone. - Safety induction before to learn proper manual handling.	- Ensure to use the help of other or lifting equipment when handling heavy objects. - Maintain a stable posture when lifting. - Ensure pathway is clear of people and objects when moving parts.	Before and during manual handling operations. (07/10/2026)	L
Operating robot when tired or distracted	Lack of concentration on robotic movement. Delayed reaction in emergency. Uploading wrong code.	- Clashes, collision causing bodily injuries. - Damage to robotic arm and surrounding environment.	- Emergency stop button is accessible - Laboratory supervision. - Induction on safety before usage.	- Do not operate robot if excessively tired and take regular breaks. - Stop operations if concentration is difficult to maintain.	Before and during robot operation. (07/10/2026)	L
Incorrect program used	Unexpected robot arm movement. Possible damage to the robot.	- Bodily injuries from clashes and collisions - Collision with objects and surrounding environment could be damaging. - Program may not be safe or have errors that could potentially damage the robot.	- Emergency stop button is accessible - Run simulations of code before uploading. - Double check/verify code before testing on the robot.	- Ensure to only have the one code file open as to not confuse it with another. - Use a unique file name for the approved code to prevent mix-up. - Maintain concentration when operating the robot.	Before the robot operation. (07/10/2026)	L
Unauthorised personal entering laboratory	No understanding of safety procedure. No PPE.	- Bodily injuries from clashes with robot. - Damaging the robot through incorrect usage.	- Ensure safety and lab inductions are completed and approved before entering lab. - Wear appropriate PPE when in lab. - Implementing Supervision. - Robot operation area is clearly visible	- Laboratory access is controlled. - Emergency stop button is accessible - Use warning signs to ensure that robot operation area is not entered.	Before and during the robot operation. (07/10/2026)	L
Collisions between robot arm and personnel	A person enters the operating area while the reBot B601-Dm robots is moving.	- Impact, pinching or crushing injuries. - Damage to the robot and end-effector.	- Emergency stop accessible. Lab supervision. - Inducted personals allowed - Restricted operation area	- Establish and clearly identify the robot's operating area. - Keep personal outside of area during operation. - Ensure all personal are aware when the robot is moving. - Use reduced speeds. - Ensure operation has ended before entering workspace.	Before and during the robot operation. (07/10/2026)	L
Pinch/crush points at robot joints and gripper	Body-parts and objects may become stuck or trapped between moving joints.	- Pinching or crushing injuries to personal. - Damage to robots, objects and surrounding environment.	- Emergency stop accessible - Supervised operation.	- Keep hand far away from robot while in operation. - End operation before retrieving or moving objects in close reach of the robots.	Before and during the robot operation. (07/10/2026)	L

Dropped or incorrectly gripped tray	Gripper may fail to securely grasp an object or the object may slip during movement.	- Falling objects could crush personal or damage equipment. - Unpredicted robot movement during an incorrect grasp.	- Use lightweight loads/objects. - Use slow speeds during testing. - Simulate code before testing to ensure correct FK and IK.	- Verify grip before executing movements. - Testing should be at low speeds and heights. - Keep personal clear from testing zone. - Ensure personal are aware of when testing begins.	During the operation and implementation of the robots (23/10/2026)	L
Collision with work cell	Robots may collide with tables, trays food objects, and other equipment due to incorrect object placement.	- Damage to robots or gripper. - Damage to surrounding environment. - Possibility of dropped objects.	- Simulation performed before physical testing. - Emergency stop button accessible. - Clear workspace of extra equipment and objects.	- Verify robot, object and target coordinate frames. - Check the completed trajectory (or whole sweep) for collisions. - Use reduced speed during first test and eliminate any unnecessary objects.	During the operation and implementation of the robots (23/10/2026)	L
Incorrect RGB camera detection	Possibility of camera incorrectly identifying tray colour.	- Incorrect meal selection. - Possibility of unpredicted robot movements - Incorrect food given to prisoners.	- Camera/program testing before operation. - Ensure there is obstruction of camera. - Try colours are bright and saturated.	- Verify detected tray classifications during initial testing. - Eliminate risk by only serving one food option per robot. Or fully eliminate by serving only one option.	During the operation and implementation of the robots (23/10/2026)	L
Multiple robot operation within shared space.	Simulated robots may have overlapping trajectories if they are coded individually.	- Collisions between robots, objects and workspace. - Damaged robots and surrounding areas.	Use collision detection to ensure the robots have a clear path and do not contact other robots or surrounding objects.	- Simulate robots and ensure they do not make contact with surroundings. - Define the individual robot operating zones. - Check completed sweep for collisions - Keep safe distance between each robot work zone.	During the operation and implementation of the robots (23/10/2026)	L
Prisoners having clear access to the robots	Escaping prison. Potential meliaceous usage. Using as a weapon	- Bodily injuries to inmates and guards. - Damaging the robot through incorrect usage.	Ensure guards are present to regulate and prevent prisoners from accessing the robots	Building a cage around the robots with security doors that prevent prisoners from accessing the kitchen with robots.	During the operation and implementation of the robots (23/10/2026)	L

Approval of assessment	I am satisfied that the residual risk with existing controls is acceptable ☒ Yes ☐ No	Signature	H.Murphy Maryam Q Y.Haddad	Date	28/09/2026
	I am satisfied that the proposed controls will reduce risk to an acceptable level. ☐ Yes ☐ No				

Guidance notes for documenting General Risk Assessments

ACTIVITY

Briefly describe this hazardous work activity - E.g. Operating, Handling, Using ... (Include names) of hazardous equipment, substances or materials used, and any quantities and concentrations of substance(s) or reaction products.

ASSOCIATED HAZARDS

Plant & Equipment – noise, vibration, moving parts (crushing, friction, stab, cut, shear), pressure vessels, lifts/hoists/cranes, sharps

Manual Handling – repetitive movements, lifting awkwardly, lifting heavy objects

Work Environment – moving objects, extremes in temperature, isolation, work at height, allergies to animal bedding, dander and fluids, risk of fire/explosion, slippery surfaces/trip hazards

People – potentially violent or volatile clients/interviewees

Communicable Diseases – exposure to bodily fluids/infectious materials, animal bites and scratches,

Environmental – emissions to atmosphere, discharge to soil and water bodies (including stormwater run-off), nuisance noise & odour, poor ventilation/air quality

Radiation (non-ionizing) – including lasers, microwaves or UV light

Electrical – plug-in equipment used in 'hostile' work environment, exposed conductors, high voltage equipment

Pathogens – dealings with pathogenic microorganisms such as bacteria, parasites, fungi or viruses

GMOs – dealings with genetically modified organisms

Cytotoxins – carcinogens, mutagens or teratogens

Radiation (ionizing) – ionizing radiation source such as radioactive substance or radionuclide, or irradiating apparatus

Chemical – hazardous substances, dangerous goods, fumes, dust, compressed gas, hazardous waste

INHERENT RISK

Provide details of the harm that could be caused to people or the environment if something goes wrong.

For example: inhalation of fumes, laceration, injury to back, infection, burns to skin or eyes.

Think about what could happen if controls fail or are not in place.

CONTROL MEASURES

Note the existing and proposed actions to reduce risk to an acceptable level. Apply the "Hierarchy of Controls", listed below, when deciding the best control measure to apply. Control types closer the top of the list are preferable.

1. ELIMINATE THE HAZARD. For example: use a different less dangerous piece of equipment, fix faulty machinery, use safer materials or chemicals
2. ISOLATE THE HAZARD FROM THE PEOPLE. Separate people from the danger. For example: use shielding, use lifting equipment or trolleys, remove dust or fumes with exhaust system, lock-out machinery.
3. CHANGE THE WAY THE JOB IS DONE. For example: change work practices, provide training, information and signs, develop work procedures.
4. USE PERSONAL PROTECTIVE EQUIPMENT (PPE), noting specific PPE is required for each job. For example: respirator, hearing protection, gloves. Training and information is required for the use of PPE.

RESIDUAL RISK LEVEL (H, M, L)

Estimate risk taking into account the way the activity is run and control measures put in place. The level of risk can be determined by combining consequence and likelihood using the risk matrix from below. Residual risk should be reduced to a level acceptable by management.

CONSEQUENCE OF HARM - This is how bad it will be if something does go wrong e.g. the number of people that could be harmed, the severity of injury.

LIKELIHOOD OF HARM - Chance of harm occurring is affected by the duration of the activity and its frequency; the number of people doing the activity and the level of exposure to the hazard.

		CONSEQUENCE					
		Insignificant	Minor	Moderate	Major	Catastrophic	
		Injury/illness consequence	Non-injury incident	Injury/ill health requiring first aid	Injury/ill health medical attention	Injury/ill health requiring hospital admission	Fatality or permanent disabling injury
		Environmental consequence	Minor effects on biological or physical environment	Moderate short term effects but not effecting ecosystem functions	Serious medium-term environmental effects	Very serious long term impairment of ecosystem functions	
LIKELIHOOD	Almost Certain	The event will occur on an annual basis	Moderate	High	High	Critical	Critical
	Likely	The event has occurred several times or more in your career	Moderate	Moderate	High	High	Critical
	Possible	The event might occur once in your career	Low	Moderate	Moderate	High	High
	Unlikely	The event does occur somewhere from time to time	Low	Low	Moderate	Moderate	High
	Rare	Heard of something like this occurring somewhere	Low	Low	Low	Moderate	Moderate

UTS SAFE WORK METHOD STATEMENT (SWMS)

1. FACULTY/SUBJECT			
Faculty/Subject title		41013 Industrial Robotics	
Subject supervisor/coordinator		Gavin Paul	
SWMS prepared by		Hannah Murphy, Maryam Qutub, Yara Haddad	

2. WORK ACTIVITY DESCRIPTION	
Describe the work activity E.g. Operating, Handling, Using.. Include names of hazardous equipment, substances or materials used, and any quantities and concentrations of substance(s) or reaction products.	LAB Activity: This activity involves the operation and handling of a reBot B601-Dm six-axis robotic arm with a gripper within the UTS laboratory environment. The physical robot will be used to demonstrate part of a proposed automated meal preparations and distribution work cell designed for a prison environment (BFED_S). The proposed work cell uses multiple robotic arms to identify, prepare and distribute meal trays according to different dietary requirements. Within this activity the real reBot B601-DM robot will perform the picking up and placing operation. This is done by picking up/grabbing the prepared food and placing it on a meal tray. The activity will involve programming and testing the robot in the UTS laboratory environment. Simulator Activity: The swift simulator will simulate a prison cafeteria environment. The simulation will include three robots that perform their own tasks. The first robot is a Hanwha HCR-5 which will scan the colour to determine which tray and food the other robots should begin to prepare. The second robot is the Elite Robots EC66 which will be in charge of preparing the food alongside the reBot B601-Dm and then place it in a predetermined end zone that will allow the prisoners to grab their food. The final robot is the Techman TM5-900. This robot will be in charge of delivering the food to the isolated prisoners. It will take the food from the kitchen and then deliver it to the to a predetermined area of the simulation where the isolated prisoners will be.

3. HAZARDS: Choose those hazard types that will need to have control measures in Section 4			
Work Environment - Working in Remote Locations - Working Outdoors/fieldwork - Clinical/Industrial setting - Poor ventilation/Air quality - Temperature extremes - Working at Height - Slip/Trip/Fall hazards	Plant - Noise - Vibration - Working with compressed air - Lifts Hoists or Cranes - Moving parts (Crushing, friction, cut, stab, shear hazards) - Pressure Vessels or Boilers	Chemical - Hazardous Chemicals use - Skin/eye irritant - Sensitiser - Mutagen - Carcinogen - Toxic to reproduction - Aquatic toxicity - Toxic - Corrosive - Dangerous when wet	Ergonomic/Manual Handling - Repetitive or awkward movements - Lifting heavy objects - Overreaching - Working above shoulder or below knee height - Poor workstation set up
Electrical - Plug in equipment - High voltage - Exposed wiring - Exposed conductors	Radiation - Ionising Radiation - Non-ionising radiation (Lasers, Microwaves, Ultraviolet light)	Biological - Sharps/Needles - Cytotoxins - Pathogens/infectious materials - Infectious materials - Communicable diseases - Animal/insects - Work with fungi/bact/viruses	Psychosocial - Aggressive or violent clients/students - Working in isolation - Working with timeframes - Staffing issues

4. CONTROLS MEASURES: Choose those that apply for hazards identified		
Eliminate/Isolate/Substitute / Engineering Controls - Remove hazard - Restrict access - Redesign equipment - Guarding / Barriers / Fume Cupboard / exhaust - Biosafety cabinet - Use safer materials/substances - Ventilation	Admin specific: Licenses/permits Work Methods - Training Information or Instruction - Licensing or certification of operators - Test and tag electrical equipment - Restricted access - Regular breaks - Task rotation - Work in pairs - Document Chemical risk assessment	Emergency Response Systems - First aid kit - Chemical spill kit - Safety shower - Eye wash station - Emergency Stop button - Remote Communication Mechanism

- Regular maintenance of equipment - Redesign of workspace / workflow	Ladder / Sling register	
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Other controls not listed

- Inspecting/checking prior to operating the robot.
- Use a safe operating speed.
- Use an exclusion zone when the robot is operating.

5. PPE REQUIRED (Tick those that apply) ☒

					
EYE PROTECTION	HEARING PROTECTION	CLOSED-IN FOOTWEAR MUST BE WORN	HAND PROTECTION	FACE SHIELD	PROTECTIVE CLOTHING
☐	☐	☒	☐	☐	☐

		
RESPIRATORY PROTECTION	LONG HAIR MUST BE CONTAINED	HEAD PROTECTION
☐	☒	☐

6. EMERGENCY EQUIPMENT

		
EMERGENCY SHOWER	EMERGENCY SPILL KIT INSIDE	EMERGENCY EYEWASH
☐	☐	☐

7. WORK ACTIVITY STEPS

BEFORE YOU START (LAB):

- Ensure all required safety documentation has been completed, reviewed and approved before operating the robot.
- Ensure all group members involved in operating the robot have completed all the required laboratory and robot safety inductions
- Ensure the required Python script is ready and prepared and that the correct file is opened and ready to be uploaded.
- Inspect the reBot B601-DM robotic arm, gripper, power supply and cables/wires for visible electrical and mechanical damage.
- Ensure the emergency stop is accessible and operational before commencing testing.
- Check the robot workspace and operating area to ensure they are clear of unnecessary objects.
- Ensure the picked up object and the end zone tray are placed in the correct spot according to the program.
- Confirm that the objects being handled are within the payload restrictions of the robot.
- Ensure all team members are wearing the correct PPE and that loose hair and fabric is secured.
- Review the planned robot trajectory before uploading.
- Ensure bystanders are aware that the robot is about to operate.
- Ensure prisoners do not have access to the robot workstation. (Simulation)
- Open the group Python project and ensure the correct and most recent version of the simulation code is being used. (Simulation)

BEFORE YOU START (SIMULATION):

- Ensure the correct and most recent version of the group python is saved and pushed to GitHub
- Check that the Hanwha HCR-5, Elite Robots EC66 and Techman TM5-900 robot models load correctly.
- Ensure all required work cell objects, including food objects, and are correctly positioned.
- Check that each robot has the correct base position and initial configuration related to it.
- Ensure the pick-up, drop-off and transfer locations are correctly defined within the work cell
- Make sure the barrier is up to prevent the prisoners from having access to the robots.

- Ensure all isolated prisoners are locked in their cells before delivering food to that section of the prison.
- Checked the planned robot trajectories for possible collisions by checking the full sweep.
- Ensure the RGB camera and colour-detection system is correctly configured to identify the colour-coded trays.

STEPS IN WORK ACTIVITY (LAB):

1. Use code to explain to the tutor the command sequence, expected robot movements, and how to stop motion in an emergency.
2. Set up the reBot B601-Dm and connect it to the controlling computer according to the required laboratory procedure.
3. Position the meal tray and food objects are the designated pick-up and end zone locations.
4. Perform an initial test run with reduced operating speeds and monitor the robot's pick-up and drop-off movements.
5. From code: Command the robot arm to move to the designated pick-up position and use the gripper to grab the food object.
6. From code: Move the object along the planned trajectory to the designated tray (end zone).
7. From code: Release the gripper and place the object into the designated tray (end zone)
8. Return the robot to its designated safe position once testing is complete.
9. Ensure program is stopped/completed before entering the robot's workspace or removing objects.
10. Remove objects belonging to the group and clean workspace to allow other groups to perform their tasks safely.

STEPS IN WORK ACTIVITY (SIMULATION):

1. Load the robots, prison environment and objects inside the swift simulation.
2. Allow the prisoners to come up to the kitchen to be scanned by the RGB and Hanwha HCR-5.
3. Command the Elite robots EC66 to pick up its designated food object and place it into the correct meal tray.
4. Add the simulated reBot B601-Dm to the environment to work with the Elite Robot EC66 to prepare the food.
5. Once meal is complete, it is transferred to the designated collection/end zone.
6. For meals designated for isolated prisoners, command the Teachman Tm5-900 to collect the completed meal and transport it to the predetermined isolation area.
7. Monitor the movements of all the robots and be ready to stop simulation in the case of a collision or error.
8. Once mealtime is over, robots return back to the initial starting safe position.
9. Stop/end the simulation.

EMERGENCY PROCEDURES:

- Press emergency button
- Notify security or dial 6 using the UTS internal phone
- Dial 000 in the case of a life threatening emergency

TRAINING REQUIRED:

- Risk Assessment
- Safety Work Method Statement (SWMS)
- Safety and Wellbeing Essentials

	Expiry date 18/01/2027	Priority 1 Safety and Wellbeing Essentials (students only)	Status COMPETENT	
• FEIT CB11 MMR Mechatronics Lab Induction				
	Expiry date 15/08/2028	Course name FEIT CB11 MMR Mechatronics Lab Induction	Status COMPETENT	

8. SIGN OFF

PREPARED BY:	LAB SUPERVISOR	DATE: 28/09/2026
NAME: HANNAH MURPHY, MARYAM QUTUB, YARA HADDAD	NAME: FELIPE ULLOA GUTIERREZ	REVIEW DATE: 02/10/2026